FALCON — Response to Q1 editorial reassessment

29 October 2026 | Internal author response to independent pre-submission review; not an official journal decision. Corrections and evidence paths refer to the accompanying 29 October frozen reviewer archive.

P0-A — Complete supplementary evidence

Status: Partially closed

The frozen reviewer archive includes all three event-window and three historical-reference collector folders, complete source-exported peer–prefix CSVs, all 12 RIB source hashes, original 2023 IODA JSON, historical 2020 IODA published-image calibration records, and Tables S1–S15. The publisher-native IODA 2020 time series and the original MRT binaries are not redistributed, and source-image reuse rights remain pending.

Evidence/location: Online Resource 1, Tables S1–S15; Evidence/BGP_2020; Evidence/Reference_2020_20200102; Evidence/IODA_2020_Original_Publisher_Image_URLs_and_SHA256.json; Code/REPRODUCE.md.

P0-B — Journal submission requirements

Status: Open — final author/publisher check

The target International Journal of Critical Infrastructure Protection (IJCIP) is within the technical scope of the paper. Its exact current article template, anonymization model, mandatory forms, figure dimensions, reference style and third-party image permissions have not been independently cleared in this turn. Publisher-general double-anonymization guidance is conditional: apply only if the target journal requires it.

Evidence/location: Review/JOURNAL_COMPLIANCE_GATE.md; author portal before submission.

P0-C — PDF metadata mismatch

Status: Closed for document title

The Word core Title and new PDF Title now equal the current scientific article title. Check author identities, title-page and data links against journal policy before final upload.

Evidence/location: Manuscript DOCX/PDF properties.

P1-A — Historical 16-hour non-event reference

Status: COMPLETE as a source-native descriptive reference-window comparison; no claim of causal identification

The 2–3 January 2020 08:00–00:00 UTC window was retrospectively selected and fixed before examining reference-window outcomes (after the event had occurred). Six original RIPE RIS RIBs for RRC00/RRC01/RRC10 were downloaded and hash-logged, decoded with the event-matching methodology, and released as complete peer–prefix CSVs. AS30873 showed reference shares of 0.48%, 0.90%, 0.41% versus event shares of 66.83%, 67.98%, 64.73%. An identical-key intersection sensitivity retained the contrast; one reference cannot establish ordinary-churn variation or cable causality.

Evidence/location: Main text Abstract, C1, §§4.3, 8.1.4, 10.6, 13; Tables S14–S15, Figure S4; reference source commit e7d0f51729ef8fffe138fa7fc34a3130455e1ad2.

P1-B — Actual neighbor transition counts

Status: Closed descriptively

Fully enumerated origin-adjacent AS_PATH transitions from the genuine frozen source-native CSV for each AS and RRC. AS30873 transition AS15412→AS174 counts 2793/1089/593 in RRC00/01/10. The 15412→197206 and 15412→6762 changes for AS12486 are now reported separately. These are conditional route observations, not evidence of physically separated cables.

Evidence/location: Results §8.1; Table S12; Figure S3; Table_S12_2020_Adjacency_Transitions_by_Collector.csv.

P1-C — Address-family and exclusion sensitivity

Status: Closed descriptively

Computed counts of first, eligible, retained, absent, unparsable and neighbor-changed observations by IPv4/IPv6. AS12486 IPv6 first rows exist, yet have zero same-origin retained comparable rows; thus its change rates use IPv4 comparables only. The reason for zero matched IPv6 observations remains unobserved, not a confirmed withdrawal.

Evidence/location: Results §8.1; Table S13; Table_S13_2020_IPv4_IPv6_Selection_and_Transitions.csv.

P1-D — Scientific contribution

Status: Improved, narrow

The manuscript already limits novelty to independently auditable, collector/peer/prefix-qualified original-RIB measurements. New directional transitions and family stratification enrich the measured finding. The evidence-to-claim checks remain reporting guidance, not a formally validated new network method.

Evidence/location: Abstract, §§3,8–9.

P2 — Figures, redundancies and DOI audit

Status: Partially closed

Figure 2 original vector and Fig 3 original SVG in bundle; new high-resolution supplemental Figure S3 added. Figure 3's deterministic range must never be labelled 95% CI. Total bibliography: 38 entries; no claim that every DOI/title/venue has been independently verified, and print-size proofing remains necessary.

Evidence/location: Figures; Supplement S3; Review/REFERENCES_38_NOT_FULLY_EXTERNALLY_VERIFIED.txt.

Verification boundaries

All original January 2020 BGP peer–prefix records, code and checksum indexes are supplied from the frozen historical source exports. The new S12/S13 tables were independently recalculated from those released CSVs. We did not download and independently decode six original RIB binary files in this reassessment, nor establish route physical causation, continuous peer session availability, raw January 2020 IODA time series, or journal acceptance likelihood.

Supplemental response: P1-A historical non-event baseline test

Reviewer concern: The observed January 2020 route reconfiguration could reflect ordinary adjacent-AS churn over any 16-hour interval. An equivalent reference window, not a 2023 signal comparison, is requested.

Response: We chose the one-week-prior 2 Jan 08:00 → 3 Jan 00:00 UTC reference window before examining its outcome, retrospectively and without prospective preregistration; retrieved the six original RIPE RIS RIBs for RRC00, RRC01, RRC10; applied the identical two-checkpoint source-native parser; checksum-logged the 12 total RIBs; exported all peer–prefix observations, equal-unit summaries, and a fixed-key inter-window comparator. Across RRC00, RRC01 and RRC10, AS30873 reference changed/comparable shares were 0.48%, 0.90%, 0.41% versus event shares 66.83%, 67.98%, 64.73%.

Locations: manuscript §4.3 (source and design), §8.1.4 (results), §10.6 (interpretation), §11 (limits); Online Resource 1 Tables S14–S15 and Figure S4; source-native reference RIB manifest and extracted record CSVs in the frozen supporting package.

Remaining inferential limit: One retrospectively selected date fixed before reference outcome inspection does not supply the distribution of ordinary BGP churn. These observations are correlated, use different prefix inventories and do not prove physical-cable causal attribution. We do not report an unsupported event-causality p value or a confidence interval.

Response to independent review of 29 October 2026

Review decision received: Targeted Major Revision before submission. The items below refer to the full 29 October 2026 corrected reviewer archive. Current numeric claims remain descriptive and noncausal.

Review item	Status	Action	Location
P0-1 — Contradictory non-event completion status in §10.6	Closed	Replaced stale non-result language with verified completion for six original reference RIBs, row-level reaggregation and explicit single-reference limitations.	§10.6; §§8.1.4,12.1; Online Resource S14–S15
P0-2a — Supplement list stops at S11	Closed	Updated Supplementary information to Tables S1–S15 and Figures S3–S4; updated data/code availability and source provenance.	Manuscript Supplementary information; Online Resource 1
P0-2b — Reproduction sequence says six RIBs only	Closed	Distinguished 6 source event RIBs and 6 source reference RIBs, provided both immutable source releases and clean-machine scripts.	§12.1; Supplement reproduction instructions; Code/REPRODUCE.md
P0-2c — Stale baseline-not-complete QA and README	Closed	Moved superseded internal QA/README to clearly labeled Historical_Superseded and regenerated release status and SHA-256 manifest.	Archive root README_FINAL_AR_EN.md; QA/FINAL_RELEASE_GATES_2026-10-29.json
P1-1 — Baseline results absent from abstract / C1 / conclusion	Closed	Explicit AS30873 reference 0.41–0.90%, event 64.73–67.98%, fixed-key restriction, single-window limits; earlier Results summary beside Table 5.	Abstract, C1, §8.1, §8.1.4, Conclusion
P1-2 — AS12486 IPv6 censoring and denominator	Closed descriptively	Added 0/54, 0/24, 0/13 event IPv6 matched-origin retention, reference 54/54, 23/23, 13/13, and explicit effectively IPv4-only neighbor-change rate	Results §8.1, Table 5 caption, Table S13

Review item	Status	Action	Location
		qualification. No IPv6 service-loss claim.	
P1-3 — One reference is not normal-churn distribution	Closed by bounded claim	Explicit single pre-outcome-selected comparator; no p-values, anomaly tail probability or causal control. Multiple non-event windows remain an optional future enhancement.	Abstract, §8.1.4, §10.6, §11, Conclusion
P1-4 — Novelty and editorial targeting	Clarified	Emphasized event-source RIB reconstruction, matched-key descriptive reference, and source-level audit; no new generic physical cable mapping method claimed.	Abstract, C1–C5, §10.5, Conclusion
P2 — S3 caption includes unchanged transitions; proofing	Closed for scoped caption	Specified unchanged neighbor pairs in Figure S3 caption, updated PDFs and metadata; future journal-specific figure and style requirements remain to be checked.	Supplement Fig S3; final PDFs
Final gate — All 38 references, third-party rights, author approval, journal policy	OPEN: requires author/editor sign-off	No claim of completed reference-by-reference external audit, third-party image permission, signed author approvals, or confirmed IJCIP anonymization/articl e-format rules.	README final submission-gate checklist

Audit scope: independent calculations on released CSV/JSON and publisher original-raster checks do not constitute a new independent re-download and parse of all twelve raw MRT files and 192 RRC00 UPDATE archives. Missing native January 2020 IODA time-series are disclosed. Standard-library smoke tests and source-hash manifests are included, with original publisher and provider rights retained.

E1 — Supplement title and release date

CLOSED: The cover now reads Tables S1–S15, 29 October 2026, and the supplement body retains the full S12–S15 analyses. PDF regenerated from the revised source DOCX.

E2 — Nonexistent paths and Atlas audit

CLOSED FOR PACKAGE INTEGRITY: The actual archive Supplement/Atlas_2020_audit.zip is supplied with its aggregated audit.json and provenance limitations. Reproduction step 4 references included executable scripts rather than missing Code/verify_archived_2020_bgp.py. The outdated Logs/BGP_2020_ARCHIVED_AUDIT_INDEPENDENT_CHECK.json reference was replaced with existing QA/BGP_rowlevel_independent_validation.json and QA/FINAL_RELEASE_GATES_2026-10-29.json. This does not claim a fresh replay of 192 original UPDATE archives.

E3 — Root README and reviewer letter QA path

CLOSED: Root README_AR.md and README_EN.md rewritten for this exact 29 October release; response table now points to QA/FINAL_RELEASE_GATES_2026-10-29.json.

E4 — Historical reference selection precision

CLOSED: The one-week-earlier date was chosen retrospectively after the event and fixed before examining the reference results. The text removes any implication of prospective preregistration or a normal-churn distribution.

E5 — Bibliography, rights and journal requirements

PARTIALLY OPEN: Original image rights are unverified and publisher-owned PNGs are not redistributed; explicit author approvals cannot be provided by an assistant. All 38 references are indexed for author/publisher metadata verification, and DOI-format tests are separated from externally verified metadata. Specific final Guide-for-Authors constraints require an official publisher sign-off; no claim of Q1 acceptance or submission clearance is made.